SHARP ASYMPTOTICS FOR QUADRATIC LAGRANGE INTERPOLATION

ABSTRACT. For distinct interpolation nodes in $[-1,1]$, let M_n denote the minimum of the integral of the sum of the squares of their Lagrange fundamental polynomials. We prove $M_n = 2 - C_*/n + o(n^{-1})$, where $C_* = 2\Gamma(5/4)^2/\Gamma(3/4)^2$. The endpoints together with the zeros of $P_{n-2}^{(5/4,5/4)}$ attain this coefficient. The lower bound follows from exact interpolation identities, compactness of an associated zero-free polynomial, and a weighted Hilbert-transform comparison. The limiting difference is a nonnegative fractional ground-state integral, with endpoint concentrations retained by lower semicontinuity.

1. INTRODUCTION

For $Y = \{y_1, \dots, y_n\} \subset [-1,1]$ with distinct elements, write $\ell_j^Y(x) = \prod_{k\neq j}(x-y_k)/(y_j-y_k)$, and set

$$I_n(Y) = \int_{-1}^{1} \sum_{j=1}^{n} \ell_j^Y(x)^2\, dx, \qquad M_n = \inf_Y I_n(Y). \tag{1}$$

The problem of determining M_n goes back to Erdős [1]. Fejér's solution of the corresponding maximum-norm problem suggested the Legendre–Gauss–Lobatto points. Szabados [2] disproved their optimality for the integral problem for every $n > 3$. Erdős, Szabados, Varma and Vértesi [3] proved $2 - O((\log n)^2/n) \le M_n \le 2 - 2/(2n-1)$. Brutman and Toledano [4] subsequently found numerical evidence against the proposed first-order coefficient 1, using optimization through $n = 100$ and extrapolation.

The discussion of problem 1131 [5] records a disproof of the coefficient-one conjecture announced by Price on 26 April 2026, followed by Sothanaphan's preliminary check on 27 April. An accessible subsequent proof and Lean development [6] use the roots of $T_n - \frac{1}{6}T_{n-2}$ and establish an eventual coefficient at least 106/105. The same discussion records reported exact small-n enclosures, and a preprint [7] whose proposed first-order formula is conditional on an endpoint universality conjecture. Those results do not supply a sharp unrestricted coefficient. In particular, the contribution here is the matching lower bound, rather than a new disproof of coefficient 1 or a claim that the numerical value had not been anticipated.

Theorem 1.1. *For the minimum in* (1),

$$\lim_{n\to\infty} n(2 - M_n) = C_* = 2\frac{\Gamma(5/4)^2}{\Gamma(3/4)^2} = 1.0942198076132386\ldots. \tag{2}$$

The arrays $X_n = \{-1,1\} \cup \{x : P_{n-2}^{(5/4,5/4)}(x) = 0\}$ *satisfy* $I_n(X_n) = 2 - C_*/n + o(n^{-1})$.

We prove a comparison which is useful independently of minimizers. Throughout the paper $I = (-1,1)$, $a(x) = 1 - x^2$, $s(x) = \sqrt{a(x)}$ and $g(x) = a(x)^{1/4}$. Integrals without indicated endpoints are over I. All continuum operators act on $L^2(I, dx)$ unless another measure is specified. The finite Hilbert transform is $Hh(x) = \pi^{-1}\,\mathrm{PV}\int h(t)/(x-t)\,dt$; functions to which $|D|$ is applied are extended by zero to $\mathbb{R}$, and $|D|h = (Hh)'$ on I. The classes $\mathcal{S}_1$ and $\mathcal{S}_2$ denote the trace-class and Hilbert–Schmidt operators. Multiplication operators are written by their multiplying functions when this causes no ambiguity.

Theorem 1.2. *If* Y_n *contains both endpoints and* $I_n(Y_n) \le 2 + C_0/n$ *for a fixed* C_0, *then* $\liminf_n n(I_n(Y_n) - I_n(X_n)) \ge 0$. *More precisely, along a subsequence attaining this lower limit there is a positive function* f *such that*

$$\liminf_n n(I_n(Y_n) - I_n(X_n)) \ge \mathcal{D}[f], \qquad \mathcal{D}[f] = \frac{1}{4\pi}\iint_{I^2} \frac{[g(x)f(y) - g(y)f(x)]^2}{f(x)f(y)(x-y)^2}\,dx\,dy < \infty. \tag{3}$$

The analytic ingredients used below are the classical Hilbert-transform theorem for Muckenhoupt A_2 weights, its reverse-Hölder consequence, the L^p boundedness of circular conjugation, and standard Hardy-space boundary theory; see [9, 10, 11]. On an interval, an A_2 weight means a positive function w for which $\sup_J(⨍_J w)(⨍_J w^{-1}) < \infty$, with J ranging over its subintervals. We use the interval versions of these theorems, obtained by reflection and extension. All problem-specific finite estimates, compactness statements and endpoint passages are proved below.

2. Jacobi interpolation and uniform reference estimates

We use the standard Jacobi normalization $P_m^{(\alpha,\alpha)}(1) = \Gamma(m+\alpha+1)/[\Gamma(\alpha+1)\Gamma(m+1)]$; see [8]. Temporarily fix $1 < \alpha < 3/2$, put $m = n-2$, $P = P_m^{(\alpha,\alpha)}$, $q = aP$, $N = m+\alpha+1/2$ and $\beta = 3/2-\alpha$. The reference points are the endpoints and the zeros x_i of P; endpoint coordinates are excluded from sums involving Gaussian weights. Let λ_i be the Gaussian weights for $a^\alpha dx$, and put

$$C_m = 2^{2\alpha+1}\frac{\Gamma(m+\alpha+1)^2}{\Gamma(m+1)\Gamma(m+2\alpha+1)}, \qquad A_\alpha = \int a^{\alpha-1}, \qquad A_m = \sum_i \frac{\lambda_i}{a_i}. \tag{4}$$

Here $a_i = a(x_i)$ and $s_i = s(x_i)$. Gaussian exactness and the Jacobi derivative identity give $\lambda_i = C_m/[a_i P'(x_i)^2]$ and $\ell_i^0(x) = q(x)/[a_i P'(x_i)(x-x_i)]$ for interior coordinates. Define $W_\alpha(x) = \mathrm{PV}\int a(t)^\alpha/(x-t)\,dt$ and $F_m(x) = \mathrm{PV}\int a(t)^\alpha P(t)/(x-t)\,dt$. These normalizations will be retained in every formula involving the second-kind function F_m.

Lemma 2.1. *Uniformly on $[-1,1]$,*

$$|P(x)| \le Cn^{-1/2}(s+n^{-1})^{-\alpha-1/2}, \qquad |F_m(x)| \le Cn^{-1/2}(s+n^{-1})^{\alpha-1/2}. \tag{5}$$

The reference angles have gaps comparable to n^{-1}, including the first and last gaps. The weights $W_i = n\lambda_i a_i^{-\alpha}$ satisfy $W_i \asymp s_i$ and

$$|W_i - \pi s_i| \le C/n, \qquad W_i = (n/N)\pi s_i + O_K(n^{-2}) \tag{6}$$

on each compact interior interval K in the second formula. The full reference square sum is bounded, and $\int(\ell_\pm^0)^2 \le Cn^{-2}$ for the endpoint cardinals.

Proof. The equations for P and F_m are respectively $aP'' - 2(\alpha+1)xP' + m(m+2\alpha+1)P = 0$ and $aF_m'' + 2(\alpha-1)xF_m' + [m(m+2\alpha+1)+2\alpha]F_m = 0$. The second follows by applying the Hilbert transform to the equation for $a^\alpha P$; that function and its first derivative vanish at the endpoints. Orthogonality identifies the Wronskian constant:

$$a(PF_m' - P'F_m) + 2\alpha xPF_m = C_m. \tag{7}$$

Both $u(\theta) = s^{\alpha+1/2}P(\cos\theta)$ and $v(\theta) = s^{1/2-\alpha}F_m(\cos\theta)$ solve $y'' + [N^2 - (\alpha^2-1/4)\csc^2\theta]y = 0$. The center values follow from the Gegenbauer generating function: if $r_j = (\alpha+1)_j/(2\alpha+1)_j$ and $\lambda = \alpha+1/2$, $P_{2k}(0) = (-1)^k r_{2k}(\lambda)_k/k!$ and $P'_{2k+1}(0) = (-1)^k 2r_{2k+1}(\lambda)_{k+1}/k!$. The other center value vanishes by parity. Equation (7) then fixes the complementary center data for F_m. Gamma-ratio estimates imply $|y(\pi/2)| + |y'(\pi/2)|/N \le Cn^{-1/2}$ for both solutions. Variation of constants and Gronwall give $|y| + |y'|/N \le Cn^{-1/2}\exp(C|\cot\theta|/N)$. Thus (5) holds when $s \ge c/n$.

For $0 \le \theta \le 1/N$, the positive ultraspherical integral representation gives $|P| \le P(1)$, $|P'| \le P'(1)$ and $P'(1)/P(1) = m(m+2\alpha+1)/[2(\alpha+1)]$. Consequently $P \asymp P(1) \asymp n^\alpha$ there. Integrating $(F_m/(Pa^\alpha))' = C_m/(a^{\alpha+1}P^2)$ from $\cos(1/N)$ proves $|F_m| \le Cn^{-\alpha}$. Reflection treats the other endpoint. Sturm comparison gives separation at least c/n. Beyond a fixed multiple of $1/n$ the equation has frequency at least $N/\sqrt{2}$; comparison there, together with the preceding endpoint layer, gives the upper gap and first-root bounds.

At a zero, (7) and (5) bound $|a_iP'(x_i)|^{-2} \le Cn^{-1}s_i^{2\alpha-1}$, whence $W_i \le Cs_i$. A local difference-quotient estimate for u gives $|u'(\theta_i)| \le C\sqrt{n}$ and therefore the reverse bound, since $W_i = nC_m s_i/u'(\theta_i)^2$. For $\mathscr{E} = u'^2 + (N^2-U)u^2$, where $U = (\alpha^2-1/4)\csc^2\theta$, one has $\mathscr{E}' = -U'u^2$. The displayed center formulas and duplication give $\mathscr{E}(\pi/2) = NC_m/\pi + O(n^{-1})$; the first-order

gamma-ratio terms cancel. Integration yields $\mathscr{E}(\theta) = NC_m/\pi + O((ns^2)^{-1})$. At a zero this is u'^2. Inverting when ns is large, and using $W_i \le Cs_i$ otherwise, proves (6).

For the remaining assertions, Gaussian partial fractions and (7) give the exact identity

$$\sum_j (\ell_j^0)^2 - 1 = \frac{B_\alpha P^2 - 2(\alpha-1)xPF_m}{C_m} + \left(\frac{A_m - A_\alpha}{C_m} + \frac{1}{2P(1)^2}\right)(1+x^2)P^2, \tag{8}$$

where $B_\alpha = A_\alpha(1+x^2-2\alpha)+2(\alpha-1)xW_\alpha$ and $A_\alpha - A_m = C_m/[2\alpha P(1)^2] = O(n^{-2\alpha})$. One way to check the latter equality is Rodrigues' formula followed by m integrations by parts in $F_m(1)$. If $G_b = H(a^b)\pi$, $b = \alpha-1$, then $B_\alpha = -a^2 G_b'$ and $G_b' = O(a^{b-1})$. Indeed for $0 < b < 1$, its Pearson equation is $aG_b' + 2bxG_b = 2bG_b(1)$, whose odd solution is $G_b(1)[x + (2b-1)a^b \int_0^x a(t)^{-b}\,dt]$. Hence $B_\alpha = O(a^\alpha)$, and every term of (8) is bounded by (5). Finally $\ell_\pm^0 = (1 \pm x)P/[2P(\pm 1)]$ and $\int P^2 = O(n^{2\alpha-2})$, also immediate from (5). This proves the endpoint mass estimate. □

Proposition 2.2. *For the reference array with parameter $1 < \alpha < 3/2$,*

$$I_n(X_n^\alpha) = 2 - \frac{c_J(\alpha)}{n} + o(n^{-1}), \qquad c_J(\alpha) = 2\frac{\Gamma(5/2-\alpha)\Gamma(\alpha)}{\Gamma(2-\alpha)\Gamma(\alpha-1/2)}. \tag{9}$$

Its largest coefficient in this parameter interval is C_*, attained at $\alpha = 5/4$.

Proof. On compact interior intervals the center-normalized differential equations in Lemma 2.1 give complementary sine and cosine expansions, with function errors $O(n^{-3/2})$. In particular nP^2/C_m converges weakly there to $1/(2\pi a^{\alpha+1/2})$, while $n\int xPF_m \to 0$. For the last assertion, in angular coordinates the integrand is $n\cos\theta\, u(\theta)v(\theta)$: it is uniformly bounded outside the $1/n$ endpoint layers, those layers contribute $O(n^{-1})$, and interior oscillations have zero mean. The last term in (8) has integral $O(n^{-2})$. Also $B_\alpha = -a[A_\alpha + (\alpha-1)W_\alpha'/\alpha]$, with W_α' bounded. Thus the endpoint majorant for $n|B_\alpha|P^2dx$ is $C\theta^{2-2\alpha}d\theta$ outside the first layer, and that layer contributes $O(n^{2\alpha-3})$. It follows that $c_J = -(2\pi)^{-1}\int B_\alpha a^{-\alpha-1/2}$.

Solving $aW_\alpha' + 2\alpha xW_\alpha = 2\alpha A_\alpha$ with $W_\alpha(0) = 0$ gives $W_\alpha' = 2\alpha A_{\alpha 2}F_1(1, 1/2-\alpha; 1/2; x^2)$. Beta integration and Gauss summation [13, 15.4.20] give

$$\begin{aligned} c_J &= \frac{B(1/2,\alpha)B(1/2,3/2-\alpha)}{2\pi}[1-4(\alpha-1)^2] \\ &= 2\frac{\Gamma(5/2-\alpha)\Gamma(\alpha)}{\Gamma(2-\alpha)\Gamma(\alpha-1/2)}. \end{aligned}$$

For example, the hypergeometric integral needed here is $\int a^{1/2-\alpha}{}_2F_1(1, 1/2-\alpha; 1/2; x^2) = 2(1-\alpha)B(1/2, 3/2-\alpha)$; both beta integrals are convergent. Writing $p = \alpha - 1/2$, $q = 2-\alpha$, one has $p+q = 3/2$ and $c_J = 2\Gamma(p+1/2)\Gamma(q+1/2)/[\Gamma(p)\Gamma(q)]$. Strict concavity of $\log\Gamma(t+1/2) - \log\Gamma(t)$, whose second derivative is $\psi_1(t+1/2) - \psi_1(t) < 0$, proves the assertion. □

The following estimates are the finite comparison input. For the interior reference mass matrix $B_{ij}^0 = \int \ell_i^0\ell_j^0$, put $\Omega_n = \mathrm{diag}(\lambda_i a_i^{-\alpha})$, $W_n = n\Omega_n$, $R_n = n(\Omega_n - B^0)$ and $\mathcal{M}_n = W_n - R_n$. Extend all these matrices by zero on endpoint coordinates. Inverses of W_n below always mean the inverse on the interior coordinates and zero on the two endpoint coordinates.

Lemma 2.3. *There are constants $\rho < 1$ and C, depending only on α, such that $0 \le R_n \le \rho W_n$, $\mathrm{tr}\, R_n \le C$, and, for every $y \in [-1,1]$,*

$$\left|\sum_i \pi s_i \ell_i^0(y)^2 - \pi s(y)\right| \le C/n, \qquad L(y)R_nL(y)^* \le C/n, \tag{10}$$

where $L(y) = (\ell_i^0(y))_i$. With $k = \alpha - 1/2$,

$$|(R_n)_{ij}| \le \frac{C}{n}\left(\frac{\min(s_i,s_j)}{\max(s_i,s_j)}\right)^k. \tag{11}$$

Proof. For $c > 1$ set $J_m(c) = \int a^\alpha P^2/(c-x)$ and $\psi_i(c) = 1/[a_i P'(x_i)(c-x_i)]$. The Gaussian resolvent error is

$$e_m(c) = \int \frac{a^\alpha}{c-x} - \sum_i \frac{\lambda_i}{c-x_i} = \frac{J_m(c)}{P(c)^2} > 0, \tag{12}$$

and the same error with a numerator $p(x)^2$, $\deg p < m$, is $p(c)^2 e_m(c)$. Both statements follow by polynomial division and Gaussian exactness. Applying $u^r = (\sin \pi r/\pi) \int_0^\infty t^{r-1} u/(u+t)\, dt$ with $r = 2-\alpha$ and $u = a(x)$, and using $(a+t)^{-1} = [(c-x)^{-1} + (c+x)^{-1}]/(2c)$, $c = \sqrt{1+t}$, gives the exact positive factorization

$$R_n = \frac{n \sin \pi(2-\alpha)}{\pi} \int_1^\infty (c^2-1)^{2-\alpha} J_m(c)[\psi(c)\psi(c)^* + \psi(-c)\psi(-c)^*]\, dc. \tag{13}$$

Positivity also justifies the integral passages by monotone convergence. Equation (5) implies $nJ_m(c) \le C/\sqrt{c^2-1}$. Together with $|a_i P'(x_i)|^{-2} \le Cn^{-1} s_i^{2k}$, this bounds the same-end part of an entry, for $u = \min(s_i, s_j)$ and $v = \max(s_i, s_j)$, by $Cn^{-1}(uv)^k \int_0^1 h^\beta/[(h+u^2)(h+v^2)]\, dh$. Splitting at u^2, v^2 bounds the integral by Cv^{-2k}. The far part is integrable since $2\beta - 2 < -1$; for opposite-end indices one denominator stays bounded near either endpoint. This proves (11), including $R_{ii} \le C/n$.

For completeness, the strict relative bound is a sampling inequality, not an assumed spectral limit. Disjoint intervals of length c/n about the reference angles have $\sin\theta \asymp s_i$. The inequality $|h(t)|^2 \le (2/|J|) \int_J |h|^2 + 2|J| \int_J |h'|^2$ and the Legendre spectral inequality $\int a|p'|^2 \le n(n-1) \int |p|^2$, $\deg p < n$, give $\sum_i s_i |p(x_i)|^2 \le Cn \int |p|^2$. Since $\Omega_i \le C s_i/n$, applying this to $p = \sum z_i \ell_i^0$ gives $B^0 \ge c\Omega_n$. Thus $R_n \le (1-c) W_n$.

We give the endpoint details of both row bounds. Put $f_0 = a^{-1/2}$, $V_\alpha = \mathrm{PV} \int a(t)^{\alpha - 1/2}/(x-t)\, dt$, and let ε_m be the Gaussian error for the divided difference $(f_0(y) - f_0(t))/(y-t)$. The Stieltjes representation $f_0(x) = \pi^{-1} \int_1^\infty (c^2-1)^{-1/2}[(c-x)^{-1} + (c+x)^{-1}]\, dc$ gives $\varepsilon_m' \ge 0$ and $\varepsilon_m' \le Cn^{-2\alpha} s^{-3}$. Indeed $e_m(c) \le e_m(1)/c$ follows by expanding in even monomials, whose Gaussian errors are nonnegative, and $e_m(1) = C_m/[2\alpha P(1)^2]$. Differentiating the cardinal Cauchy transform and using (7) now gives exactly

$$\sum_i \pi s_i \ell_i^0(y)^2 = \pi s + \frac{\pi}{C_m}[(1-2\alpha) y s P F_m - a^2 P^2 V_\alpha' - a^2 P^2 \varepsilon_m']. \tag{14}$$

The Pearson equation used above, with exponent $\alpha - 1/2$, gives $|V_\alpha'| \le C a^{\alpha - 3/2}$. Consequently the first two corrections are $O(n^{-1})$ by (5). The last is bounded by $Cn^{-2\alpha} s P^2 \le C/n$, using separately $s \ge n^{-1}$ and $s \le n^{-1}$. Continuity includes the endpoints.

For the second row bound in (10), set $b = \alpha - 1$, $G_b = \mathrm{PV} \int a^b/(y-t)\, dt$ and $H_m(c) = \sum_i \lambda_i/[a_i(c-x_i)]$. Partial fractions give the exact exterior interpolant

$$q_c(y) := L(y)\psi(c) = \frac{aP[G_b(y) - H_m(c)] - F_m(y) + y(A_\alpha - A_m)P(y)}{C_m(c-y)}. \tag{15}$$

Gaussian positivity in the even-power expansion gives $0 \le G_b(c) - H_m(c) \le Cn^{-2b}$ for $c \ge 1$. To verify the rate, differentiate (12) at 1: if $\Lambda = m(m+2\alpha+1)$ and $D_m = A_\alpha - A_m$, then $e_m'(1) = -D_m[\alpha\Lambda/(\alpha^2-1) + \alpha/(\alpha-1)]$; the largest error is $\int a^{\alpha-2} - \sum_i \lambda_i/a_i^2 = (D_m - e_m'(1))/2$. Also $|G_b(y) - G_b(c)| \le C(s^{2b} + (c-1)^b)$ for $y \ge 0$, $1 < c < 2$, by the same Pearson formula. The second-kind equation implies $|F_m'| \le Cn^{2-\alpha}$ when $s \le n^{-1}$: integrate $(a^{-b} F_m')' = -(\Lambda + 2\alpha) a^{-b-1} F_m$ from $s \asymp n^{-1}$, where the bound follows from the transformed equation. Since $F_m(1) = D_m P(1)$, it follows that $|F_m - y D_m P| \le C s^2 n^{2-\alpha}$ in that layer. Writing $h = c - 1$, (15) therefore yields

$$|q_c(y)| \le Cn^{-1/2} \frac{s^{b+1/2} + s^\beta h^b}{h + s^2} \qquad (s \ge n^{-1},\ 0 < h < 1),$$

$$|q_c(y)| \le C \frac{s^2}{h+s^2}(n^{1-b} + n^{1+b} h^b) \qquad (s \le n^{-1},\ 0 < h < 1). \tag{16}$$

For q_{-c}, $y \ge 0$, and also for q_c when $c \ge 2$, the bounds are $Cn^{-1/2}/c$ and Cn^{b-1}/c in the respective ranges. For the first line of (16), scaling $h = s^2u$ gives

$$s^{2b+1}\int_0^1 \frac{h^\beta}{(h+s^2)^2}\,dh \le C, \qquad s^{2\beta}\int_0^1 \frac{h^{\beta+2b}}{(h+s^2)^2}\,dh \le C.$$

For the second line, split at $h = n^{-2}$; its bounds become Cn^{1-b} below that point and $Cn^{b-1}h^{b-1}$ above it. Their squared integrals against $h^\beta dh$ are $O(n^{-1})$, because $b+\beta = 1/2$. The opposite-end and far-exterior integrals are also $O(n^{-1})$, since $\int_1^\infty (c^2-1)^\beta c^{-2}\,dc < \infty$. Insertion into (13) proves $L(y)R_nL(y)^* \le C/n$. Reflection completes all cases. □

3. Finite comparison and polynomial bounds

From now on $\alpha = 5/4$ and $X_n = X_n^{5/4}$, although estimates whose proofs work for $1 < \alpha < 3/2$ retain that parameter. For another endpoint-containing array put $A_{ij} = \ell_j^0(y_i)$, $J_n = A^*A - I$, $U_n = A^* - A^{-1}$ and $V_n = W_n^{1/2}U_n$. The endpoint rows of A are the corresponding rows of the identity. A change of the cardinal basis gives the following exact formula, with the indicated estimate for the omitted mass blocks:

(17)
$$n(I_n(Y) - I_n(X_n)) = -\operatorname{tr}(W_nJ_n) + \operatorname{tr}(R_nJ_n) + \operatorname{tr}(\mathcal{M}_nU_nU_n^*) + \epsilon_{n,e}, \qquad |\epsilon_{n,e}| \le Cn^{-1/2}\,\|V_n\|_{\mathcal{S}_2}.$$

Indeed $U_nU_n^* = J_n + A^{-1}A^{-*} - I$, and the interior mass is $(W_n - R_n)/n$. For the endpoint estimate, positivity of the full old mass matrix gives $(B_{IE}^0)^*(B_{II}^0)^{-1}B_{IE}^0 \le B_{EE}^0$. Since $B_{II}^0 \le W_n/n$, Lemma 2.1 yields $\left\|W_n^{-1/2}nB_{IE}^0\right\|_{\mathcal{S}_2} \le Cn^{-1/2}$. If $E = A - I$ and $G = A^{-1} - I$, then $Ge_\pm = -U_ne_\pm$; the endpoint diagonal block of $A^{-1}A^{-*} - I$ is zero. Cauchy–Schwarz proves (17) without an unweighted inverse bound.

Let K_n be the reproducing kernel of polynomials of degree less than n in $L^2(dx)$, and let $\chi_n(x) = K_n(x,x)^{-1}$. The following uniform estimate will be used also near the endpoints:

$$\chi_n(\cos\theta) = \pi\sin\theta/n + O(n^{-2}), \qquad 0 \le \theta \le \pi. \tag{18}$$

Here is an elementary verification. Christoffel–Darboux gives $2K_n(x,x) = n^2P_n(x)^2 + aP_n'(x)^2$, where $P_n(1) = 1$. Put $u = \sqrt{\sin\theta}P_n(\cos\theta)$ and $N_0 = n+1/2$. Then $u'' + [N_0^2 + (4\sin^2\theta)^{-1}]u = 0$, so for $E = u^2 + (u'/N_0)^2$, $|E'| \le E/(4N_0\sin^2\theta)$. The center binomial formulas give $E(\pi/2) = 2/(\pi N_0) + O(N_0^{-2})$. Integration gives $E = 2/(\pi N_0) + O((N_0^2 s)^{-1})$ when $\theta \in [1/n, \pi - 1/n]$. Since $2sK_n = n^2u^2 + (u' - \frac{1}{2}\cot\theta\, u)^2$, this implies $K_n = n/(\pi s) + O(s^{-2})$. Invert when ns exceeds a sufficiently large constant. In the remaining endpoint range, Markov's inequality $|P_j(\cos\theta) - 1| \le j^2\theta^2/2$ gives $K_n \ge cn^2$ by summing $j \le c'n$; both terms of (18) are then $O(n^{-2})$.

Lemma 3.1. *Every distinct array satisfies $I_n(Y) \ge 2 - C/n$. If it contains the endpoints and $I_n(Y) \le 2 + C_0/n$, then*

(19)
$$\|V_n\|_{\mathcal{S}_2} \le C, \quad \pi\sum_i s(y_i) = 2n + O(1), \quad \mathscr{X}_n := I_n(Y) - \sum_i \chi_n(y_i) = O(n^{-1}), \quad \|B_Y\| \le C/n,$$

where $(B_Y)_{ij} = \int \ell_i^Y \ell_j^Y$.

Proof. The residual $r_i = \ell_i^Y - \chi_n(y_i)K_n(\cdot, y_i)$ is orthogonal to the comparison kernel, so $\int(\ell_i^Y)^2 = \chi_n(y_i) + \|r_i\|_2^2$. Consequently (18) gives $\pi\sum_i s(y_i) \le nI_n(Y) + C$. By (10), the uniform reference square-sum bound and (6),

$$\operatorname{tr}(W_nJ_n) = \pi\sum_i s(y_i) - 2n + O(1) \le n(I_n(Y) - 2) + C. \tag{20}$$

The reference sum is $2n/\pi + O(1)$: the transformed Jacobi equation gives $N\theta_i = \pi i + O(1)$ uniformly, by integrating its phase equation from $\theta \asymp n^{-1}$, and summing the Lipschitz function $\sin\theta$ gives the assertion.

By (10), every row of $E = A - I$ has $E_i R_n E_i^* \le C/n$. Hence $\left\| R_n^{1/2} E^* \right\|_{\mathcal{S}_2} \le C$ and, writing $\mathcal{P} = \|V_n\|_{\mathcal{S}_2}^2$, $\left\| R_n^{1/2} G \right\|_{\mathcal{S}_2} \le C + \sqrt{\rho \mathcal{P}}$. As $A^{-1}A^{-*} - I = G + G^* + GG^*$, $|\operatorname{tr}\{R_n(A^{-1}A^{-*} - I)\}| \le \rho\mathcal{P} + C\sqrt{\mathcal{P}} + C$. Using this and (20) in the equivalent form of (17) gives

$$2n(I_n(Y) - 2) \ge n(I_n(X_n) - 2) + (1 - \rho)\mathcal{P} - C\sqrt{\mathcal{P}} - C. \tag{21}$$

Proposition 2.2 and completion of the square prove the lower bound and the first assertion of (19) for endpoint-containing arrays. The finite identity then also bounds $\operatorname{tr}(W_n J_n)$ below, yielding the sine sum and the excess bound.

For the global mass estimate let $\Lambda = \operatorname{diag}(\chi_n(y_i))$ and $G_{ij} = \sqrt{\chi_n(y_i)\chi_n(y_j)}K_n(y_i, y_j)$. Then $G_{ii} = 1$, $B_Y = \Lambda^{1/2} G^{-1} \Lambda^{1/2}$, and $P = B_Y + \Lambda^{1/2} G \Lambda^{1/2} - 2\Lambda \succeq 0$ has $\operatorname{tr} P = \mathscr{X}_n$. Thus $B_Y \le 2\Lambda + \mathscr{X}_n I \le CI/n$.

To remove the endpoint hypothesis in the lower bound, write $S = \sum \ell_i^2$. At either boundary outside the node hull the extreme cardinal has modulus at least one, and therefore $S(1), S(-1) \ge 1$. Under dilation of the nodes about a point c of their hull the derivative at scale one is $I_n - (1 - c)S(1) - (1 + c)S(-1) < 0$ if $I_n < 2$. This sign persists throughout admissible dilation. Expand until one endpoint is reached, then expand about that endpoint until the other is reached. The integral decreases, proving the unrestricted bound. The case $n = 1$ is immediate. □

For later use we record further consequences with their justifications. Normalize the nodal polynomial Q_n to leading coefficient 2^{n-1}. Then

$$c \le \|Q_n\|_2 \le C, \qquad n \sum_j Q_n'(y_j)^{-2} \le C. \tag{22}$$

Indeed $Q_n = T_n - \mathcal{I}_Y T_n$, and the mass bound in (19) controls the interpolant of the bounded nodal data $T_n(y_j)$. The lower bound is the least norm for a prescribed leading coefficient, attained by the Legendre polynomial; its squared norm tends to $\pi/4$. To prove the second assertion, put $Z_n = \sum_j \int (x - y_j)^2 \ell_j^Y(x)^2 \, dx = \|Q_n\|_2^2 \sum_j Q_n'(y_j)^{-2}$. The residual decomposition above and Christoffel–Darboux give $Z_n \le C/n + 8\mathscr{X}_n$: the comparison term for a single node is bounded by $C\chi_n(y_j)^2[p_{n-1}(y_j)^2 + p_n(y_j)^2] \le C/n^2$, where $p_j = \sqrt{j + 1/2} P_j$ and $p_{n-1}(\cos\theta)^2 + p_n(\cos\theta)^2 \le C/(s + n^{-1})$. The latter follows from the same differential-equation bound used in (18); in the first layer use $|P_j| \le 1$.

Bounded objective implies that the empirical angular measures tend to $d\theta/\pi$. To see this without a spacing assumption, let E be the orthonormal-polynomial evaluation matrix. Since $I_n = \operatorname{tr}((E^*E)^{-1}) \le C$, the arithmetic–geometric mean inequality gives $\det(E^*E) \ge (n/C)^n$. Factoring the determinant into leading coefficients and the Vandermonde product gives

$$\liminf_{n \to \infty} \frac{2}{n^2} \sum_{i<j} \log|y_i - y_j| \ge -\log 2.$$

For any weak limit μ of the empirical measures, truncating the logarithmic kernel below gives logarithmic energy at least $-\log 2$. The cosine expansion $\log|\cos\theta - \cos\phi| = -\log 2 - 2\sum_{k \ge 1} \cos(k\theta)\cos(k\phi)/k$, read by Abel regularization, shows that the energy is at most $-\log 2$, with equality only if every nonconstant cosine moment vanishes. Thus the limit is the arcsine measure. This proves the assertion and uniform convergence of the ordered angular quantiles to the uniform quantile map.

Lemma 3.2. *On each compact subinterval of $(0, \pi)$ the ordered new angles $\phi_i = \arccos y_i$ have two-sided gaps comparable to n^{-1}. Their normalized kernel Gram matrices satisfy $c_K I \le G_K \le C_K I$ and $\|G_K - I\|_{\mathcal{S}_2} \le C_K$.*

Proof. The excess identity is $\mathscr{X}_n = \sum_i \chi_n(y_i)[(G^{-1})_{ii} - 1]$. On a compact interior interval $\chi_n \ge c_K/n$, and Schur complement inversion gives $(G^{-1})_{KK} \ge G_K^{-1}$. Since $\operatorname{tr} G_K$ equals its dimension, its eigenvalues μ_j obey $\sum_j (\mu_j - 1)^2/\mu_j \le C_K$. This gives the two spectral bounds and the Hilbert–Schmidt bound. The unit kernel vector $u_n(\theta) = (p_j(\cos\theta))_{j<n}/\sqrt{K_n(\cos\theta, \cos\theta)}$

has derivative norm at most $C_K n$. The spectral lower bound on a two-node principal matrix gives $\|u_n(\phi_i) - u_n(\phi_j)\|^2 \geq 2c_K$, proving separation.

For the upper bound, a gap of length H/n with midpoint t in a smaller fixed interval admits the nonnegative polynomial $p_t(\cos\theta) = F_m(\theta - t) + F_m(\theta + t)$, where here $m = \lfloor n/2 \rfloor$ and $F_m(u) = m^{-1}[\sin(mu/2)/\sin(u/2)]^2$ is the Fejér kernel. It has degree less than n, $\|p_t\|_2 \geq c\sqrt{n}$, and $0 \leq p_t(\cos\theta) \leq Cn/(1+n|\theta-t|)^2$. Separation inside a larger compact interval and the bound $O(1/n)$ per sample outside it give $\sum_i p_t(y_i) \leq C[n/(1+H)+1]$. On the other hand, synthesis and $\left\|\ell_i^Y\right\|_2 \leq C/\sqrt{n}$ give $\|p_t\|_2 \leq Cn^{-1/2}\sum_i p_t(y_i)$. It follows that $c \leq C/(1+H) + C/n$, so H is bounded. □

4. Boundary convergence of the associated polynomials

Define analytic polynomials in the disk by

$$F_n(z) = 2z^n Q_n((z+z^{-1})/2), \qquad b_n(z) = F_n(z) - zF_n'(z)/(2n). \tag{23}$$

They have constant coefficient one, and

$$b_n(e^{i\theta}) = e^{in\theta}[Q_n(\cos\theta) - isQ_n'(\cos\theta)/n], \qquad f_n(x) := |b_n|^2 = Q_n^2 + aQ_n'^2/n^2. \tag{24}$$

All zeros of F_n are on the unit circle, counted with multiplicity. The function $H_n = 1 - zF_n'/(nF_n) = (2n)^{-1}\sum_{\xi:F_n(\xi)=0}(\xi+z)/(\xi-z)$ has positive real part in the disk and $b_n = F_n(1+H_n)/2$. Thus b_n is zero-free there. Its possible boundary zeros are the simple zeros at ± 1 corresponding to endpoint nodes.

We will use the exact reciprocal identity

$$\int \frac{s}{f_n}\,dx = \pi n \sum_j Q_n'(y_j)^{-2} \leq C. \tag{25}$$

Here is a derivation, including the endpoint justification. First move all nodes into the open interval. Since $\deg b_n < 2n$ and it is zero-free on the closed disk, the identity $Q_n/f_n = \frac{1}{2}[z^n/b_n(z) + z^{-n}/b_n(z^{-1})]$ on the circle shows by Fourier orthogonality that Q_n is the degree-n orthogonal polynomial for $dx/(sf_n)$. The same Fourier calculation, or integration of the Cauchy kernel on the circle, gives

$$\int \frac{Q_n(t)}{z-t}\frac{dt}{s(t)f_n(t)} = \frac{\pi w(z)^n}{\sqrt{z^2-1}\,b_n(w(z))}, \qquad w(z) = z - \sqrt{z^2-1}, \tag{26}$$

where the square root is asymptotic to z at infinity. For clarity, expanding the Cauchy kernel uses $(z-\cos\theta)^{-1} = (1 + 2\sum_{k\geq 1} w^k \cos k\theta)/\sqrt{z^2-1}$; only the positive frequencies $z^n/b_n(z)$ remain in the integral. At a zero y_j, the second-kind quotient and (26) give the Gaussian weight $\lambda_j = \pi/[nf_n(y_j)] = \pi n/[a(y_j)Q_n'(y_j)^2]$. Gaussian exactness on the polynomial a proves (25). If an endpoint is a node, move it inward by ε at fixed n. With t its distance from that endpoint, the denominator obeys $f_{n,\varepsilon} \geq c_n(t+\varepsilon^2)$: the derivative term supplies t, and the polynomial value supplies ε^2 when t is smaller. Hence $s/f_{n,\varepsilon} \leq C_n t^{-1/2}$. Dominated convergence proves the endpoint version. The bound is now (22).

The Legendre derivative inequality and (22) imply $\int_0^\pi s|b_n|^2 d\theta \leq C$; (25) gives $\int_0^\pi s^2|b_n|^{-2}d\theta \leq C$. Since $\log^p(1+t) \leq C_p t$ and $\log\sin\theta$ belongs to every finite L^p, the positive and negative parts of $u_n = \log|b_n|$ are uniformly bounded in every finite angular L^p. The normalized analytic logarithms $\log b_n(0) = 0$ therefore have bounded H^p norms for $1 < p < \infty$, by circular conjugation. Also $(1-z^2)b_n$ is bounded in H^2. After extraction F_n converges locally uniformly: the generating function gives $|z^j p_j((z+z^{-1})/2)| \leq C_r$ on $|z| \leq r < 1$; Cauchy–Schwarz in the Legendre coefficients of Q_n, followed by $\sum_{j\leq n} r^{2(n-j)} \leq C_r$, bounds F_n on each smaller disk. The derivative term in (23) tends locally to zero, so b_n has the same local limit F. Hurwitz's theorem gives a zero-free F with $F(0) = 1$. The logarithms converge locally to $\log F$ and weakly in H^p. In particular

$$\log F(z) = \frac{1}{2\pi}\int_0^{2\pi} \frac{e^{it}+z}{e^{it}-z}\log|F(e^{it})|\,dt, \qquad \fint_{\mathbb{T}} \log|F| = 0. \tag{27}$$

Thus F is outer; this also excludes singular logarithmic terms at the endpoints before any boundary compactness is used.

Lemma 4.1. *On every compact interior interval the weights f_n and f_n^{-1} have uniformly bounded A_2 characteristics and uniformly bounded $L^{1+\varepsilon}$ norms for some $\varepsilon > 0$.*

Proof. Write $d_i = Q_n'(y_i)$. Complex sampling at separated interior points $t_j + ih/n$ gives, for $\deg p < n$, $\sum_j |p(t_j + ih/n)|^2 \le Cn \|p\|_2^2$. Indeed orthogonality of associated Legendre polynomials gives $\int a^r |p^{(r)}|^2 \le (2n)^{2r} \|p\|_2^2$. The point-evaluation inequality on disjoint intervals of length c/n therefore bounds the derivative samples by $Cn(C_K n)^{2r} \|p\|_2^2$. Taylor expansion and the ℓ^2 triangle inequality sum these estimates with factors $(h/n)^r/r!$. The mass bound in (19) implies that the matrix $C_{ji} = \ell_i^Y(y_j + i\epsilon)$, $\epsilon = h/n$, is bounded on every interior square block. Since $|Q_n(y_j + i\epsilon)/(i\epsilon d_j)|^2 = \prod_{k\neq j}[1 + \epsilon^2/(y_j - y_k)^2]$ is bounded above and below on such rows, the matrix $D_{ji} = i\epsilon d_j/[d_i(y_j - y_i + i\epsilon)]$ is bounded too. For a consecutive block J and $Y = \operatorname{diag}(y_j)$, $[Y, D] + i\epsilon D = i\epsilon(d_j/d_i)_{j,i\in J}$. Taking the rank-one norm and subtracting the midpoint from Y yields

$$\left(\sum_{i\in J} |d_i|^2\right)\left(\sum_{i\in J} |d_i|^{-2}\right) \le C_K |J|^2, \tag{28}$$

because the block diameter is at most $C_K|J|/n$. In particular neighboring derivative magnitudes have bounded ratios.

On a consecutive cell $[r, t]$, let $\Delta = t - r$ and $L(x) = \log|Q_n(x)/[(x-r)(x-t)]|$. Then $|L''| \le C_K n^2$, while its two endpoint values differ by a bounded amount by (28). Thus $|L'| \le C_K n$ and $|Q_n/[(x-r)(x-t)]| \asymp |d_i|/\Delta$ there. Writing $u = (x-r)/\Delta$, the two terms in (24) are controlled by $u(1-u)$ and $1 - 2u + \Delta L' u(1-u)$ respectively. The first is bounded below away from the cell ends, and the second is bounded below near those ends. Consequently $f_n \asymp_K |d_i|^2/n^2$ on the entire cell. Comparison with cellwise constant weights transfers (28) to continuous A_2; partial cells use the neighboring ratio bound. The weighted L^1 bound for f_n, Jensen's inequality and the global logarithmic bounds control the means of both weights on fixed intervals. The reverse-Hölder theorem for A_2 weights [10] proves the last assertion. □

Lemma 4.2. *After extraction, $b_n \to F$ and $b_n^{-1} \to F^{-1}$ strongly in L^2 on compact interior arcs. Their normalized logarithms converge strongly in every finite angular L^p on the whole circle. If $\gamma_0(\theta) = \frac{1}{4}(\theta - \pi/2)$ and $\delta = \operatorname{Im}\log F - \gamma_0$, then the step functions with values $n(\phi_i - \theta_i)$ converge locally strongly in every finite L^p to δ.*

Proof. Christoffel–Darboux and the center-normalized Legendre equation give, uniformly on every compact interior angular square, $G_{ij} = \operatorname{sinc}(n(\phi_i - \phi_j)) + O_K(n^{-1})$. This follows directly by substituting the complementary sine and cosine expansions of P_n, P_{n-1} in the kernel formula; the difference between $(2\sin((\phi_i - \phi_j)/2))^{-1}$ and $(\phi_i - \phi_j)^{-1}$ is bounded, and the diagonal is obtained by differentiation. In particular, for $h_i = n(\phi_{i+1} - \phi_i)$, Lemma 3.2 gives $c \le h_i \le C$ and $\sum_i \operatorname{sinc}^2 h_i \le C$. Choose a nearest positive integer multiple πk_i except for the bounded number of gaps near zero. Then $\sum_i |h_i - \pi k_i|^2 \le C$ and $\sum_i |h_i - \pi k_i| = O(\sqrt{n})$. Equidistribution on each fixed interval gives $\sum_i h_i - \pi\#\{i\} = o(n)$, so $\sum_i (k_i - 1) = o(n)$ and

$$n^{-1} \sum_i |h_i - \pi|^2 \longrightarrow 0. \tag{29}$$

Thus every fixed number of neighboring gaps approaches the π spacing outside a set of indices of density tending to zero.

Fix such interior roots y_n and write $s_n = s(y_n)$. Define $P_n(t) = nQ_n(y_n + s_n t/n)/(s_n Q_n'(y_n))$. Its scaled roots tend to πj, $j \in \mathbb{Z}$, and $P_n(0) = 0$, $P_n'(0) = 1$. The logarithmic curvature after removing the central and next root is uniformly bounded on that cell. Neighboring derivative ratios, from (28), then bound $c_n = P_n''(0)/2 = -\sum_{j\neq 0} r_{j,n}^{-1}$. Separation gives $\sum_{|r_{j,n}|>R} |r_{j,n}|^{-2} \le C/R + C/n$. The exact canonical product $P_n(t) = te^{c_n t} \prod_{j\neq 0}(1 - t/r_{j,n}) e^{t/r_{j,n}}$ therefore tends, after extraction, to $e^{ct}\sin t$ locally uniformly with derivatives. Applying (28) on each fixed

block $-M \le j \le M$ gives $(\sum e^{2c\pi j})(\sum e^{-2c\pi j}) \le C(2M+1)^2$. Letting M increase forces $c = 0$. A diagonal argument in the window size and the tolerance, using (29), shows that this sine approximation holds on all but a vanishing fraction of the interior cells.

For $\theta = \phi_i + u/n$ in such a cell, the brackets in (24), divided by $s(y_i)Q_n'(y_i)/n$, are $\sin(-u) - i\cos(-u) + o(1) = -ie^{-iu} + o(1)$ uniformly. Their modulus is bounded below. Hence $\sigma_n = b_n/\overline{b_n}$ differs in local L^2 by $o(1)$ from the step function with values $-e^{2in\phi_i}$. The full Gram bound also gives

$$\sum_{i \ne j} \frac{|e^{2in\phi_i} - e^{2in\phi_j}|^2}{(i-j)^2} \le C_K. \tag{30}$$

Indeed multiply the sinc kernel entries by $n(\phi_i - \phi_j)/(i-j)$ and use the two gap bounds; the squared $O(1/n)$ errors sum to $O(1)$. Piecewise linear interpolation on the uniform index grid is bounded in $H^{1/2}$ by (30); pairs of adjacent cells use the adjacent differences, already included in that sum. Its squared L^2 distance from the step function is at most $Cn^{-1}\sum_i |e^{2in\phi_{i+1}} - e^{2in\phi_i}|^2 = o(1)$. Compactness gives a strong local L^2 limit σ, with $|\sigma| = 1$ and $\sigma \in H^{1/2}_{\mathrm{loc}}$. The quantile maps between the index grid and the actual cells are uniformly bi-Lipschitz locally and tend uniformly to the identity. Composition preserves the strong limit, first for continuous test functions and then by L^2 approximation. Thus $\sigma_n \to \sigma$ locally strongly.

Set $B_n = (1-z^2)b_n$. Weak H^2 convergence gives weak local L^2 convergence of b_n to F; F is nonzero almost everywhere by (27). Passing to $b_n = \sigma_n \overline{b_n}$ shows $\sigma = F/\overline{F}$. We spell out the upgrade to strong amplitude convergence. For a smooth cutoff χ in an interior arc let $\tau = (1-e^{2i\theta})\sigma/\overline{1-e^{2i\theta}}$ and $h_n = B_n - (1-z^2)F$. Reverse-Hölder in Lemma 4.1 and strong convergence of the bounded quotients imply $\chi h_n = \tau\chi\overline{h_n} + o_{L^2}(1)$. Here h_n is weakly null in H^2 and has zero constant coefficient. Let P_+ be projection onto the nonnegative circular Fourier modes, and $P_- = I - P_+$. A smooth cutoff extension τ_e of τ belongs to $L^\infty \cap H^{1/2}$ globally. Its commutator with P_+ is Hilbert–Schmidt, because its squared matrix sum is $\sum_k |k||\widehat{\tau_e}(k)|^2$. Therefore

$$\begin{aligned} P_-(\chi h_n) &= [P_-, \chi]h_n \longrightarrow 0, \\ P_+(\chi h_n) &= [P_+, \tau_e]\chi\overline{h_n} + \tau_e[P_+, \chi]\overline{h_n} + o_{L^2}(1) \longrightarrow 0. \end{aligned}$$

Both commutators are compact and their inputs are weakly null. This proves strong local L^2 convergence of b_n. Convergence in measure, nonvanishing of F, and the uniform $L^{2+\varepsilon}$ bound for b_n^{-1} prove the reciprocal convergence by uniform integrability. The same reasoning with $|\log t|^{p+1} \le C_p(t + t^{-1})$ proves local logarithmic convergence in every finite L^p. The global bounds at a larger exponent make the endpoint logarithmic tails uniformly small. Thus the real parts of the logarithms converge globally in every finite L^p; circular conjugation gives the same statement for their imaginary parts.

It remains to fix the real displacement rather than its value modulo π. Normalize $\gamma_n = \operatorname{Im}\log b_n$ by $\log b_n(0) = 0$. On (ϕ_i, ϕ_{i+1}), with $q_n(\theta) = Q_n(\cos\theta)$, the positive-real function H_n above gives exactly

$$\gamma_n(\theta) = n\theta - \pi i + \arctan\frac{q_n'(\theta)}{nq_n(\theta)}, \qquad \gamma_n(\phi_i) = n\phi_i - \pi(i - 1/2). \tag{31}$$

The arctangent is its principal value; the normalized logarithm fixes the displayed branch. On the cells with the sine approximation the continuous argument differs uniformly by $o(1)$ from its left nodal value. On the other cells that difference is bounded by the upper gap constant plus π, so their total contribution tends to zero in every finite local L^p. For the reference nodes, $n\theta_i - \pi(i-1/2) = \gamma_0(\theta_i) + O_K(n^{-1})$, by the root-anchored Jacobi expansion. Subtracting this identity from (31) and transferring quantile grids proves the last assertion of the lemma. □

We specify the embeddings used for the remaining matrix limits. Order both arrays in increasing angle, starting with the endpoint 1, and conjugate their matrices by $\epsilon_i = (-1)^{i+1}$, the signs of the corresponding nodal derivatives. Map the ith coordinate to a constant of height $\sqrt{n/\pi}$ on $((i-1)\pi/n, i\pi/n)$, and then apply the unitary map $h(\theta) \mapsto h(\arccos x)/\sqrt{s(x)}$

to physical L^2. Extend A_n, A_n^{-1} by the identity off the step space, and positive finite mass operators by zero. These conventions preserve every finite trace and norm identity. In particular $D_n = W_n^{1/2}$ tends strongly to $D = \sqrt{\pi}\, g$.

We shall need convergence at both nodal grids. If p_n has trigonometric degree at most $C_1 n$ and t_j are c/n separated in $K \Subset K_1$, then, for every $M \geq 3$,

$$\frac{1}{n} \sum_{t_j \in K} |p_n(t_j)|^2 \leq C \int_{K_1} |p_n|^2 + C_{K,K_1,M} n^{1-M} \int_{\mathbb{T}} |p_n|^2. \tag{32}$$

To prove it, reproduce p_n with a smooth Fourier cutoff kernel V_n, for which $|V_n(t)| \leq C_M n (1 + n \operatorname{dist}(t, 2\pi\mathbb{Z}))^{-M}$ and $\|V_n\|_1 \leq C_M$. Weighted Cauchy–Schwarz in the convolution and separation bound the sum of the resulting decay kernels by a constant, and by Cn^{1-M} outside K_1. If analytic polynomials P_n are bounded in H^2, have degree $O(n)$ and converge strongly locally in L^2, their step samples therefore converge strongly locally to the same limit: subtract a fixed polynomial approximation to the weak H^2 limit in (32), let n increase and then improve the approximation. Apply this to $(1 - z^2) b_n$ and divide by $1 - e^{2i\theta}$ on interior arcs. On both the reference and new grids the samples of b_n converge strongly locally to F. In particular, with $c = \cos\delta$, $p = \sin\delta$,

$$\operatorname{Step}_Y \frac{|Q_n'(y_i)|}{n} \longrightarrow \frac{|F|}{s}, \quad \operatorname{Step}_X(\epsilon_i Q_n(x_i)) \longrightarrow |F| p, \quad \operatorname{Step}_X \frac{\epsilon_i Q_n'(x_i)}{n} \longrightarrow \frac{|F| c}{s}. \tag{33}$$

The first formula uses (24) at a zero; the other two use the reference root phase in (31).

Proposition 4.3. *Put $f = 2^{-1/2} |F(e^{i \arccos x})|^2$, $T = g^{1/2} H g^{-1/2}$ and $T_f = f^{1/2} H f^{-1/2}$. The operators $D_n A_n^{-1}$ converge weakly to a unique bounded operator B. The product $L = T_f D$ has a bounded extension, and*

$$B = cD + pL, \qquad TB = cL - pD. \tag{34}$$

Proof. The finite algebra gives $\|A_n D_n h\|^2 + \|A_n^{-*} D_n h\|^2 = 2\|D_n h\|^2 + \|U_n^* D_n h\|^2$. Thus $A_n D_n$ and $D_n A_n^{-1}$ are uniformly bounded. For $h \in C_c^\infty(I)$ the cardinal formula gives exactly

$$\sum_j \ell_j^Y(x) \frac{Q_n'(y_j)}{n} h(y_j) = \frac{Q_n'(x)}{n} h(x) - \frac{Q_n(x)}{n} \sum_j \frac{h(x) - h(y_j)}{x - y_j}. \tag{35}$$

The divided difference is continuous on the closed square. Equidistribution implies uniform convergence of its normalized sum on compact sets to $-H(h/s)$; the Hilbert transform of the arcsine density vanishes in I. Give the new grid the raw vector $d_{j,n} = \sqrt{\pi/n}\, Q_n'(y_j) h(y_j)/n$. In the physical embedding it tends strongly to $k_h = |F| h / s^{3/2}$, by (33). Sampling (35) at the old nodes and multiplying by D_n gives locally strongly

$$B k_h = \sqrt{\pi}\, |F| \{ ch/s + pH(h/s) \}. \tag{36}$$

This is a global L^2 function: $H(h/s)$ is bounded near the endpoints and $\int |F|^2 dx \leq C$ by the weighted amplitude bound. The span of k_h is dense in L^2 since its multiplying function is positive almost everywhere and locally square integrable. Uniform boundedness identifies all weak operator subsequential limits on this dense family, proving uniqueness and convergence.

The second identity needs an endpoint check. Define on $\mathbb{C} \setminus [-1, 1]$ $\mathcal{R}(z) = F(w(z))/(1 - w(z)^2)^{1/4}$. Its upper boundary value is $re^{-i\delta}$, where $r = \sqrt{f/g}$, and $\mathcal{R}(\infty) = 1$. Multiply $\mathcal{R}$ by the Cauchy transform of h/s. Its real and imaginary boundary parts are $r[cH(h/s) - ph/s]$ and $-r[ch/s + pH(h/s)]$. There is no endpoint residue: from $(1 - z^2) F \in H^2$, the integral on a physical circle of radius ε at either endpoint is bounded by $C\varepsilon^{1/8} \int_{-\pi}^{\pi} |\cos(t/2)|^{-1/2}\, dt = o(1)$. The compactly supported Cauchy datum is bounded there, and the product decays at infinity. Cauchy's formula therefore gives

$$TB k_h = \sqrt{\pi}\, |F| \{ cH(h/s) - ph/s \}. \tag{37}$$

The fixed transform T is bounded: its power exponents are within the A_2 range. Rotating (36) and (37) yields $cB - pTB = D$ and $pB + cTB = T_f D$ on the dense family. The latter defines

the bounded extension L, and rotating back proves (34). Only the product $T_f D$ is asserted bounded. □

5. The global limiting comparison

Define $T_0 = g^{-1/2} H g^{1/2} = -T^*$ and $A = c - pT_0 = c + pT^*$. These are bounded operators since the fixed power weights belong to A_2. The reference limit is

$$R_n \longrightarrow R = \tfrac{1}{2} D(I - T^*T)D \quad \text{in } \mathcal{S}_1, \qquad K := TD = \sqrt{\pi}\, g^{1/2} H g^{1/2}, \quad K^* = -K. \tag{38}$$

Here and in what follows finite matrices are embedded as specified above. We give the global convergence details, because trace convergence on compact interior squares alone would be insufficient.

In (13) let $v_n(\zeta)$ be the embedded, sign-modulated vector $-\sqrt{C_m}\psi(\zeta)$. This sign makes the interior nodal derivatives agree with the positive leading coefficient of Q_n. Root derivative asymptotics give $v_n(\zeta) \to v(\zeta) = s^{\alpha-1}/(\zeta - x)$ strongly, for each fixed $\zeta > 1$. The endpoint tails follow from $|v_{n,i}(\zeta)|^2 \le C n^{-1} s_i^{2\alpha-1}/(\zeta - x_i)^2$ and the reference separation. Also $nJ_m(\zeta)/C_m \to 1/(2\sqrt{\zeta^2 - 1})$ by the interior oscillatory average of $na^\alpha P^2/C_m$ and its integrable endpoint bound. Uniformly in n, $\|v_n(\zeta)\|^2 \le C(\zeta - 1)^{\alpha-2}$ for $1 < \zeta < 2$ and $\|v_n(\zeta)\|^2 \le C\zeta^{-2}$ for $\zeta \ge 2$. The first bound follows by splitting the angular sum at $\theta \asymp \sqrt{\zeta - 1}$, with the first angular point at c'/n. Thus the trace norm of the factorization integrand is bounded by $C(\zeta - 1)^{-1/2}$ near 1 and $C\zeta^{2\beta-2}$ at infinity. Rank-one convergence and dominated convergence in $\mathcal{S}_1$ prove the first assertion of (38).

For identification, put $V_\beta(x) = \mathrm{PV} \int a(t)^\beta/(x - t)\, dt$. Its Pearson equation, or the odd power series at zero, gives $V_\beta(x) = \cos(\pi\beta) \int_1^\infty (\zeta^2 - 1)^\beta [(\zeta - x)^{-1} - (\zeta + x)^{-1}]\, d\zeta$. The limiting kernel from (13) is therefore

$$R(x, y) = \frac{g(x)^{1/2} g(y)^{1/2}}{2\pi} \frac{V_{1/4}(x) - V_{1/4}(y)}{x - y}. \tag{39}$$

The finite Hilbert-transform identity $HM_hH + M_h = [M_{Hh}, H]$, with zero extension and initially smooth compact data, identifies this kernel with $\frac{1}{2}D(I - T^*T)D$. Extension by boundedness gives (38); in particular $R = \frac{1}{2}(D^2 + K^2)$.

We also have the weighted convergences

$$D_n^{-1} R_n \to D^{-1} R \quad \text{in } \mathcal{S}_2, \qquad D_n^{-1} R_n D_n^{-1} \to \tfrac{1}{2}(I - T^*T) \quad \text{strongly}. \tag{40}$$

To verify the endpoint assertions, (11) and $2k > 1$ give $\sum_j |R_{n,ij}|^2 \le C s_i/n$. For the projection P_ε onto old coordinates with $s_i \le \varepsilon$, it follows that $\|P_\varepsilon D_n^{-1} R_n\|_{\mathcal{S}_2}^2 \le C(\varepsilon + n^{-1})$. For the other orientation use $R_n W_n^{-1} R_n \le \rho R_n$ and $R_{n,ii} \le C/n$ to obtain $\|D_n^{-1} R_n P_\varepsilon\|_{\mathcal{S}_2}^2 \le C(\varepsilon + n^{-1})$. Interior kernel convergence proves the first statement of (40). For the second, test first on compactly supported functions, control the output tails by the preceding estimates, and use $0 \le D_n^{-1} R_n D_n^{-1} \le \rho I$ and density. These arguments also give $RD^{-1} \in \mathcal{S}_2$.

The weighted forward maps satisfy

$$A_n D_n \longrightarrow AD \quad \text{strongly}. \tag{41}$$

Here is a dense-family proof which also controls their endpoint rows. The vectors $\xi_n(\zeta) = D_n^{-1} v_n(\zeta)$ converge strongly to $\xi(\zeta) = s^{-1/4}/[\sqrt{\pi}(\zeta - x)]$. Indeed their discrete squared coordinates are exactly $n^{-1} a_i^{\alpha-1}/(\zeta - x_i)^2$; separation controls the tails. The span of $\xi(\zeta)$, $\zeta > 1$, is dense, since expansion in ζ^{-1} reduces orthogonality to all polynomial moments against the positive multiplier $s^{-1/4}$. Substitute the common-phase expansions of P, F_m in the exact interpolant (15). Lemma 4.2 then gives $A_n v_n(\zeta) \to Av(\zeta)$ locally strongly; its two terms are $v(\zeta)c$ and $-pT_0 v(\zeta)$, by partial fractions in $G_{\alpha-1}$. This calculation is valid for unbounded displacement values: the error from replacing $(N/n)\delta_{i,n}$ by $\delta_{i,n}$ has local mean square tending to zero, since the latter step functions converge locally in L^2. For each fixed $\zeta > 1$, (15) and (5) give $\sup_y |q_\zeta(y)| \le C_\zeta n^{-1/2}$. Each endpoint row thus has squared magnitude at most C_ζ/n; equidistribution bounds the number of rows in an angular strip of width ε by $O(n\varepsilon) + o(n)$.

This upgrades the vector convergence to global strong convergence. Uniform boundedness of A_nD_n, proved in Proposition 4.3, and density prove (41).

Consequently the weak Hilbert–Schmidt limit of V_n is $V = DA^* - B$, and the positive term in (17) obeys

$$\liminf_n \operatorname{tr}(\mathcal{M}_n U_n U_n^*) \geq \tfrac{1}{2}(\|V\|_{\mathcal{S}_2}^2 + \|TV\|_{\mathcal{S}_2}^2). \tag{42}$$

In fact $V_n = (A_nD_n)^* - D_nA_n^{-1}$ is bounded in $\mathcal{S}_2$, and its matrix coefficients are identified by (41) and Proposition 4.3. The left side is $\left\|(I - D_n^{-1}R_nD_n^{-1})^{1/2}V_n\right\|_{\mathcal{S}_2}^2$; strong convergence of the positive square roots and weak lower semicontinuity prove (42). Thus no endpoint concentration has been removed from this term.

The compact trace has an actual limit:

$$\operatorname{tr}(R_nJ_n) \longrightarrow \operatorname{tr}\{R(A^*A - I)\}. \tag{43}$$

On a compact exterior-parameter interval this follows from the strong vector convergence just proved in (13). Its sampled parameter tails are uniformly negligible. To check this, split new rows into a fixed interior interval and its complement. For interior rows (15) gives $|q_{\pm\zeta}(y)| \leq C_\varepsilon/(\sqrt{n}\,\zeta)$, $\zeta \geq 1$; the resulting scalar majorant is $C_\varepsilon(\zeta^2-1)^\beta/\zeta^2$, integrable at both ends. For each other row the whole factorization contributes at most C/n by (10); their total limiting contribution is $O(\varepsilon)$ by equidistribution. First remove the parameter tails with ε fixed and then let $\varepsilon \downarrow 0$. The old trace tails have their independent trace-norm majorant. This proves (43) without a bound for A_n itself.

We next identify the signed trace, keeping its real linear term. Define

$$\mathfrak{a}(\delta) = \int_0^\pi \cos\theta\,\delta(\theta)\,d\theta, \quad Z(\delta) = \frac{1}{\pi}\int_0^\pi [\pi(1-2\alpha)x\sin\delta\cos\delta - s^{3-2\alpha}V_\alpha'(x)\sin^2\delta]\,d\theta, \quad \alpha = 5/4. \tag{44}$$

The coefficients in Z are bounded, including the endpoints. We claim

$$\limsup_n \operatorname{tr}(W_nJ_n) \leq \mathfrak{a}(\delta) + Z(\delta). \tag{45}$$

Put $W_{0,n} = \operatorname{diag}(\pi s_i)$ and $Z_n(y) = n[\sum_i \pi s_i \ell_i^0(y)^2 - \pi s(y)]$. The first bound in (10) gives $\sup|Z_n| \leq C$. On every compact interior interval (14) and the common-phase expansions give $Z_n(\cos\theta) = \mathcal{Z}(x, e^{2id_n(\theta)}) + o(1)$, where

$$\mathcal{Z}(x,z) = \tfrac{\pi}{2}(1-2\alpha)x\operatorname{Im} z - \tfrac{1}{2}s^{3-2\alpha}V_\alpha'(x)(1-\operatorname{Re} z), \qquad d_n(\theta) = N\theta - \tfrac{\pi}{2}(\alpha - 1/2). \tag{46}$$

For example $nPF_m/C_m = \sin d_n \cos d_n/s + O_K(n^{-1})$. Also $nP^2/C_m = \sin^2 d_n/(\pi s^{2\alpha+1}) + O_K(n^{-1})$; the Gaussian error term tends to zero uniformly in the interior. At new nodes Lemma 4.2 gives $e^{2id_n(\phi_i)} \to e^{2i\delta}$ in step L^2. Since Z_n is uniformly bounded and the endpoint row proportion tends to zero with the strip width, it follows that $n^{-1}\sum_i Z_n(y_i) \to Z(\delta)$.

Replacing $W_{0,n}$ by W_n costs $o(1)$ in this trace, not just $O(1)$. To see the needed precision set $e_i = W_i - (n/N)\pi s_i$. By (6), $|e_i| \leq C/n$ globally and $|e_i| \leq C_K/n^2$ in the interior. The row sum $\sum_i e_i\ell_i^0(y)^2$ is $O(1/n)$ globally. For interior y, split old columns into a larger interior interval and its complement; the former contributes $O_K(n^{-2})$. The latter does too, since the cardinal quotient and $\sum_i |q'(x_i)|^{-2} = O(1)$ give an exterior column square sum $O_K(n^{-1})$. The same endpoint split as above proves $\sum_{\text{new}}\sum_i e_i\ell_i^0(y_j)^2 = o(1)$ and $\sum_{\text{old}} e_i = o(1)$. Finally $\operatorname{tr}(W_{0,n}J_n) = O(1)$ by (19) and (10), and $n/N = 1 + O(1/n)$. We have proved

$$\operatorname{tr}(W_nJ_n) = \pi\left[\sum_i s(y_i) - \sum_i s(x_i)\right] + n^{-1}\sum_i Z_n(y_i) + o(1). \tag{47}$$

For the remaining linear moment let Q_n^0 have the reference zeros and the same leading coefficient as Q_n, and put $\mathcal{R}_n = Q_n/Q_n^0$. An exact Stieltjes integral, followed by integration by parts, gives

$$\pi\left[\sum_i s(y_i) - \sum_i s(x_i)\right] = \int_0^\infty \log|\mathcal{R}_n(\sqrt{1+t^2})\mathcal{R}_n(-\sqrt{1+t^2})|\,dt. \tag{48}$$

Indeed insert $a^{1/2} = \pi^{-1}\int_0^\infty u^{-1/2}a/(a+u)\,du$; the equal node counts cancel its constant term, and the logarithmic derivative of the product supplies the remaining resolvents. At the lower endpoint the boundary term is $\sqrt{c^2-1}\log|\mathcal{R}_n(c)\mathcal{R}_n(-c)| \to 0$; at infinity the odd Laurent coefficients cancel.

Only an upper bound on the positive endpoint tail of the integrand is needed. Write $\Delta = \sqrt{c^2-1}$ and $\varphi = c+\Delta$. The Legendre integral representation and (22) give $|Q_n(c)|+|Q_n(-c)| \le C\varphi^n \min(n, \Delta^{-1})$ for $1 < c \le 2$. The reference has the lower bound $|Q_n^0(c)| \ge c_\alpha \varphi^n \Delta^2/(\Delta + n^{-1})^{2-\beta}$. For a direct verification of this last bound, write $Q_n^0 = 2m!(x^2-1)C_m^{\alpha+1/2}(x)/(\alpha+1/2)_m$ and use its nonnegative coefficient expansion in $r = \varphi^{-2}$. Restrict the convolution sum to $j \le \min(m/2, c/(1-r))$. The coefficient ratio is bounded below, and the partial binomial sum is at least $c\min(m, (1-r)^{-1})^{\alpha+1/2}$, which is the asserted bound. Consequently, for $h_n(c) = \log|\mathcal{R}_n(c)\mathcal{R}_n(-c)|$,

$$[h_n(c)]_+ \le C + 2(1+\beta)\log(1/\Delta) \qquad (n^{-1} \le \Delta \le 1),$$
$$[h_n(c)]_+ \le C + 2(1+\beta)\log n + 4\log(1/(n\Delta)) \qquad (0 < \Delta < n^{-1}).$$

Since $c\,dc/\Delta = d\Delta$, the limiting positive tail on $0 < t < \varepsilon$ is at most $C\varepsilon(1+|\log\varepsilon|)$. At infinity $|h_n(c)| \le C/c^2$ uniformly: the normalized analytic polynomials F_n are bounded and zero-free on a fixed small disk, and the sum of their logarithms at w and $-w$ has vanishing linear term. Local uniform convergence, reverse Fatou with the preceding positive majorants, and (48) yield

$$\limsup_n \pi\left[\sum_i s(y_i) - \sum_i s(x_i)\right] \le \int_0^\infty \log|\mathcal{R}(\sqrt{1+t^2})\mathcal{R}(-\sqrt{1+t^2})|\,dt. \tag{49}$$

The normalized logarithm has the Cauchy representation $\log\mathcal{R}(z) = \pi^{-1}\int \delta(x)/(z-x)\,dx$. This follows from its upper boundary imaginary part $-\delta$; the endpoint circles vanish because $\log F \in H^p$, $p > 1$, and the value at infinity is zero. The integral in (49) is therefore exactly $\int x\delta(x)/s(x)\,dx = \mathfrak{a}(\delta)$: the inner t integral of $\pi^{-1}[(\sqrt{1+t^2}-x)^{-1} - (\sqrt{1+t^2}+x)^{-1}]$ is $x/s(x)$. Fubini is absolute since δ is in angular L^1. Together with (47) this proves (45).

Combining (17), (42), (43) and (45) proves the fully global reduction

$$\begin{gathered}\liminf_n n(I_n(Y_n) - I_n(X_n)) \ge \mathcal{E}(F),\\ \mathcal{E}(F) = -\mathfrak{a}(\delta) - Z(\delta) + \operatorname{tr}\{R(A^*A - I)\} + \tfrac{1}{2}(\|V\|_{\mathcal{S}_2}^2 + \|TV\|_{\mathcal{S}_2}^2).\end{gathered} \tag{50}$$

Every term here is finite. Possible negative endpoint concentration in (48) improves the inequality; positive concentration in (42) has been retained.

6. The fractional comparison identity

We first prove finiteness of the nonlinear integral; it is not an assumption in the approximation argument.

Lemma 6.1. *For the limit supplied above, $L - K \in \mathcal{S}_2$ and $\mathcal{D}[f] = \frac{1}{4}\|L + L^*\|_{\mathcal{S}_2}^2 < \infty$.*

Proof. The fixed operator $E_0 = DT - K = [D, K]D^{-1}$ is Hilbert–Schmidt. Its kernel is $\pi^{-1/2}\sqrt{g(x)/g(y)}[g(x) - g(y)]/(x-y)$. To check the endpoints put $u = 1-x$, $v = 1-y$ and $q = 1/4$. On $u \le v/2$ its square is bounded by $Cu^q v^{q-2}$; in the opposite orientation it is bounded by $Cu^{3q-2}v^{-q}$. Integration reduces both to $C\int_0^1 t^{2q-1}dt < \infty$. Comparable distances use the mean value theorem and give the same radial integral. Opposite endpoints have a bounded denominator and integrable power factors.

By (34), $V = DTp - pL$ and $KT = 2RD^{-1} - D$. Therefore the two operators

$$Kp - pL = V - E_0 p, \qquad Kc - cL = TV - 2RD^{-1}p \tag{51}$$

are Hilbert–Schmidt. Set $r = \sqrt{f/g}$. On separated compact rectangles the kernel of L is $r(x)K(x,y)/r(y)$ by its defining formula in Proposition 4.3. The squared kernels of the two operators in (51) add to

$$|K(x,y)|^2[1 + t^2 - 2t\cos(\delta(x) - \delta(y))] \ge |K(x,y)|^2(t-1)^2, \qquad t = r(x)/r(y) > 0. \tag{52}$$

A countable union of separated rectangles covers the complement of the diagonal. Hence the difference kernel $K(x,y)(r(x)/r(y)-1)$ is square integrable globally. It equals $L-K$ as an operator: on compact intervals the local A_2 realization makes the truncated Hilbert formulas converge on the dense data family of Proposition 4.3; their difference kernels converge in Hilbert–Schmidt norm. The two limits agree there, and then globally by exhaustion and boundedness. This also rules out a remaining diagonal multiplication operator. Since $K^* = -K$, the symmetric part of L has kernel $[f(x)g(y) - g(x)f(y)]/[\sqrt{\pi}\sqrt{f(x)f(y)}(x-y)]$. Its squared Hilbert–Schmidt norm is exactly $4\mathcal{D}[f]$. □

For the approximation define

$$\|h\|_X^2 = \int \frac{|h|^2}{s}\,dx + \mathfrak{e}(h), \qquad \mathfrak{e}(h) = \iint_{I^2} \frac{g(x)g(y)|h(x)-h(y)|^2}{(x-y)^2}\,dx\,dy. \tag{53}$$

We use $h(\cos\theta)$ as an even circular function and write Jh for its odd circular conjugate, restricted to the upper semicircle. Then $Jh = -sH(h/s)$ and $H(Jh) = h - \fint_0^\pi h(\cos\theta)d\theta$. These identities follow on Chebyshev polynomials from $JT_k = sU_{k-1}$ and $H(sU_{k-1}) = T_k$, and then by density.

Lemma 6.2. *Both J and H are bounded on X. If h is in angular L^2 and the kernel $Q_h(x,y) = K(x,y)(e^{h(x)-h(y)} - 1)$ is Hilbert–Schmidt, there are real $h_j \in C_c^\infty(I)$ such that $h_j \to h$ in X and $Q_{h_j} \to Q_h$ in $\mathcal{S}_2$.*

Proof. Set $x = \tanh t$ and $a_0(t) = \cosh(t)^{-1/2}$. Since $d\theta = a_0(t)^2 dt$, direct substitution gives

$$\mathfrak{e}(h) = \iint_{\mathbb{R}^2} \frac{a_0(t)a_0(u)|h(t)-h(u)|^2}{\sinh^2(t-u)}\,dt\,du, \qquad \|h\|_X \asymp \|a_0 h\|_{H^{1/2}(\mathbb{R})}. \tag{54}$$

Here is a verification of the equivalence. For $v = a_0 h$, expansion of the square gives the translation-invariant $1/\sinh^2$ form of v, plus $2\int |v(t)|^2 b(t)dt$, where $b(t) = \mathrm{PV}\int [a_0(u)/a_0(t) - 1]/\sinh^2(t-u)\,du$. The function b is bounded: subtract the odd linear Taylor term near $u = t$, and use $a_0(u)/a_0(t) \le e^{|u-t|/2}$ on the tails. The reverse expansion has the same bound. The translation-invariant form plus L^2 is equivalent to $H^{1/2}$ because its kernel is comparable to $|t-u|^{-2}$ near zero and has integrable tails.

The convolution operator $\mathscr{H}$ with principal-value kernel $1/(\pi\sinh t)$ is bounded on $H^{1/2}(\mathbb{R})$, since its Fourier multiplier is $-i\tanh(\pi\xi/2)$. The exact transformed formulas are $J = -\mathscr{H}$ and $H = a_0^{-2}\mathscr{H}a_0^2$. After the isometry $h \mapsto a_0 h$, they become $-a_0\mathscr{H}a_0^{-1}$ and $a_0^{-1}\mathscr{H}a_0$. Their differences from $\mp\mathscr{H}$ have kernels $\pi^{-1}[a_0(t)/a_0(u) - 1]/\sinh(t-u)$, or the inverse ratio. These kernels and their first output derivatives have uniform integrable Schur envelopes. At the diagonal the linear zero removes the singularity, including after differentiation by cancellation of the leading terms; on the tails the bound is $Ce^{-|t-u|/2}$. Thus the differences map L^2 into H^1. This proves the asserted bounds on X by (54).

Symmetry and $z^2 \le C[(e^z-1)^2 + (e^{-z}-1)^2]$ give $\mathfrak{e}(h) \le C\,\|Q_h\|_{\mathcal{S}_2}^2$. Clip h at $\pm N$. Clipping preserves the sign and decreases the magnitude of every difference, so $|e^{h_N(x)-h_N(y)} - 1| \le |e^{h(x)-h(y)} - 1|$. Dominated convergence gives both $Q_{h_N} \to Q_h$ in $\mathcal{S}_2$ and $h_N \to h$ in X. For fixed N, multiply by a smooth hyperbolic cutoff χ_R equal to one on $[-R,R]$ and zero outside $[-R-1, R+1]$. The capacity estimate $\mathfrak{e}(1-\chi_R) + \|1-\chi_R\|_{L^2(d\theta)}^2 \le Ce^{-R}$ follows from (54), since $a_0(1-\chi_R)$ and its derivative have squared L^2 norms $O(e^{-R})$. In the difference of $(1-\chi_R)h_N$, the part containing $h_N(x) - h_N(y)$ is a vanishing tail of its finite energy; the other part is bounded by $N^2\mathfrak{e}(\chi_R)$. Thus these cutoffs converge in X. Their uniform boundedness makes the exponential Lipschitz on the relevant difference range and gives convergence of Q in $\mathcal{S}_2$ as well. Ordinary interior mollification preserves the bound and converges in $H^{1/2}$ on compact sets, hence in X and in Q. A diagonal choice proves the lemma. □

We now evaluate the finite expression (50). First suppose $\delta \in C_c^\infty(I)$ is real, put $\eta = H\delta$, $r = e^\eta$ and $f = gr^2$. All ratios f/g are smooth and positive on the closed interval. The normalized Cauchy factor $\mathcal{R}(z) = \exp\{\pi^{-1}\int \delta(t)/(z-t)\,dt\}$ has boundary value $re^{-i\delta}$. Its

inverse times a Cauchy datum gives, by Plemelj's formula, $HM_{r-1}(c-pH) = M_{r-1}(p+cH)$. Consequently, if $A_g = c - pT$ and $C_g = p + cT$, then

$$C_g = T_f A_g, \qquad A_g^{-1} = c + pT_f, \qquad A_g D = DA, \qquad DA^{-1} = A_g^{-1} D. \tag{55}$$

For the inverse assertion, $cA_g + pC_g = I$ gives a left inverse. Both A_g and A are Fredholm of index zero: extended by the identity outside I, they differ by trace-class operators from $c - pH_{\mathbb{R}}$, whose parametrix is $c + pH_{\mathbb{R}}$ modulo smooth Hilbert-transform commutators. This applies along the homotopy $t\delta$, starting from the identity. Thus the left inverse of A_g is its inverse. The bounded identity $A_g D = DA$ then proves injectivity of A, and index zero proves its invertibility. No unbounded similarity by D^{-1} has been used.

Here are the trace details, including the trace of a commutator which cannot be suppressed. Let $U = A^* - A^{-1}$, $J = A^*A - I$, $\mathsf{V} = A^{-1}$, $\mathsf{F} = \mathsf{V}\mathsf{V}^* - I$ and $\mathsf{M} = D^2 - R$. All the differences whose traces occur below are trace class. This can be checked by expanding the smooth divided-difference kernels in a Fourier basis: two derivatives in the compact row or column variable make the sum of rank-one trace norms finite. The separated endpoint pieces have square integrable power factors. In particular $\mathsf{V} = c + pH_{\mathbb{R}} + \mathcal{S}_1$. For $k = a^{1-\alpha}$ the diagonal calculation for J, using $HM_hH + M_h = [M_{Hh}, H]$, gives

$$\operatorname{tr}(D^2 J) = \int (1-2\alpha)\frac{x}{s} cp\, dx - \frac{1}{\pi}\int kp^2 V_\alpha'\, dx = Z(\delta). \tag{56}$$

For precision, the cross-term diagonal is $-[(cp)' + (k'/k)cp]/\pi$ and the quadratic-term diagonal is $-(H(kp^2))'/(\pi k)$. Integration by parts against πs and the self-adjointness of $|D|$ give (56). Since $UU^* = J + \mathsf{F}$, (50) becomes $\mathcal{E} = \operatorname{tr}(\mathsf{M}\mathsf{F}) - \mathfrak{a}(\delta)$.

The required correction is

$$\operatorname{tr}(\mathsf{V}^*\mathsf{M}\mathsf{V} - \mathsf{M}) - \operatorname{tr}\{\mathsf{M}(\mathsf{V}\mathsf{V}^* - I)\} = -\mathfrak{a}(\delta). \tag{57}$$

Indeed the difference is the trace of $[\mathsf{V}^*, \mathsf{M}\mathsf{V}]$. The trace-class corrections R and $\mathsf{V} - (c + pH_{\mathbb{R}})$ change this trace only by commutators having a trace-class factor, whose traces vanish. Write $W = \pi s$ and extend it smoothly to the line near the compact phase support. The change outside that support contributes a trace-class separated kernel with zero diagonal and zero trace. For $[c - H_{\mathbb{R}}p, W(c + pH_{\mathbb{R}})]$ the cross-term diagonal integrates to

$$\frac{1}{\pi}\int [Wpc' + (Wc)'p]\, dx = -\frac{1}{\pi}\int W\delta'\, dx = -\mathfrak{a}(\delta).$$

The quadratic term is $Q = -H_{\mathbb{R}}(Wp^2)H_{\mathbb{R}} - Wp^2$. It is trace class and anticommutes with $H_{\mathbb{R}}$; conjugation by the invertible $H_{\mathbb{R}}$ changes Q to $-Q$, so $\operatorname{tr} Q = 0$. This proves (57) by actual traces of differences, rather than cyclically rearranging divergent traces. It follows that $\mathcal{E} = \operatorname{tr}(\mathsf{V}^*\mathsf{M}\mathsf{V} - \mathsf{M})$.

The identities $A_g^*A_g + C_g^*C_g = I + T^*T$ and (55) now imply

$$\mathcal{E} = \tfrac{1}{2}\operatorname{tr}\{D(T_f^*T_f - T^*T)D\} = \mathcal{C}[g] - \mathcal{C}[f], \qquad \mathcal{C}[h] = \tfrac{1}{2}\int \frac{s}{h}|D|h\, dx. \tag{58}$$

To verify the last equality without an infinite trace subtraction, the kernel identity above gives $\mathcal{C}[h] = \frac{1}{2}\operatorname{tr}\{D(I - T_h^*T_h)D\}$. For $h = g$ this is the trace of R. For $h = f$, the difference $T_f - T$ has a smooth divided-difference kernel times square integrable endpoint powers, hence is trace class by the same separated expansion. The diagonal of the resulting difference is $s|D|h/h$. Endpoint cutoff limits are justified by the power $1/4$.

The fractional ground-state identity evaluates (58) explicitly. Direct symmetrization of $|D|h(x) = \pi^{-1}\,\mathrm{PV}\int_{\mathbb{R}}[h(x) - h(y)]/(x-y)^2\, dy$ gives

$$\mathcal{C}[g] - \mathcal{C}[f] = \frac{1}{4\pi}\iint_{I^2} \frac{[g(x)f(y) - g(y)f(x)]^2}{f(x)f(y)(x-y)^2}\, dx\, dy = \tfrac{1}{4}\,\|L + L^*\|_{\mathcal{S}_2}^2\,. \tag{59}$$

In fact the contribution of $\mathbb{R}\setminus I$ to each $\mathcal{C}[h]$ is $\pi^{-1}\int s^{-1} = 1$, so those contributions cancel. The remaining numerator is $s(x)f(y)/f(x) + s(y)f(x)/f(y) - 2g(x)g(y)$. All integrals are finite

for a smooth positive ratio f/g. This is the usual quadratic fractional ground-state representation, also commonly expressed as a Picone identity; see [12]. Its elementary expansion here fixes its normalization. For an independent evaluation of the constant, $\mathcal{C}[g] = \frac{1}{2}\langle g, |D|g\rangle$ and $\widehat{g}(t) = \sqrt{\pi}\Gamma(5/4)(2/|t|)^{3/4}J_{3/4}(|t|)$, with $\widehat{h}(t) = \int e^{-itx}h(x)dx$. The beta-integral expansion of the exponential proves this transform. The Weber–Schafheitlin formula [13, 10.22.57] gives $\int_0^\infty t^{-1/2}J_{3/4}(t)^2dt = \sqrt{2}/\Gamma(3/4)^2$. Its convergence conditions hold both at zero and infinity. Plancherel therefore gives $\mathcal{C}[g] = 2\Gamma(5/4)^2/\Gamma(3/4)^2 = C_*$, in agreement with Proposition 2.2.

Finally return to the possibly rough limit F. Set $\eta = \log r$, so η and δ are the real and imaginary parts of $\log[F(z)/(1-z^2)^{1/4}]$ and $\delta = J\eta$. They are in every finite angular L^p. Lemma 6.1 supplies the nonlinear kernel hypothesis of Lemma 6.2. Choose its approximants η_j and put $\delta_j = J\eta_j$, $L_j = K+Q_{\eta_j}$, $p_j = \sin\delta_j$, $c_j = \cos\delta_j$. Bounded conjugation gives $\delta_j \to \delta$ in X. The sine and cosine compositions also converge in X: their fractional differences have uniformly integrable squares by strong X convergence, and their Lipschitz constants are at most one. Thus $[K, p_j] \to [K, p]$ in $\mathcal{S}_2$, and likewise for c_j. The identity $V = E_0 p + [K, p] - p(L-K)$ shows that $V_j \to V$ and $TV_j \to TV$ in $\mathcal{S}_2$. Multiplication by c_j, p_j converges strongly, as do its adjoints, so $\operatorname{tr}\{R(A_j^*A_j - I)\}$ converges. The two signed integrals converge by angular L^2 convergence. Every term of the finite expression (50) is therefore continuous in this approximation, as is $\frac{1}{4}\left\|L_j + L_j^*\right\|_{\mathcal{S}_2}^2$.

For each fixed smooth compact modulus η_j, its conjugate has the form $\delta_j = s\psi_j$ with $\psi_j \in C^\infty([-1,1])$; this follows either from the Cauchy formula or its rapidly decreasing sine series. Cut this phase off in physical endpoint strips of width ε. The cutoff tends to δ_j in X by the capacity estimate in Lemma 6.2. Its finite Hilbert conjugate tends to $\eta_j - \fint \eta_j$ in X, and also uniformly with error $O_j(\sqrt{\varepsilon})$. To check the uniform estimate, rescale $H[(1-\chi_\varepsilon)s\psi_j]$ at each endpoint: it is $\sqrt{\varepsilon}$ times the Hilbert transform of a uniformly bounded $C^{1/2}$ function with square-root vanishing at one end and smooth vanishing at the other. Subtracting its value in the principal value leaves the integrable kernel $C|u-v|^{-1/2}$; the boundary logarithm is bounded by the square-root factor. Uniform boundedness and the exponential Lipschitz estimate give convergence of the nonlinear Hilbert–Schmidt kernels as well. Hence the compact-phase calculation (59) applies first to this cutoff, then to (η_j, δ_j), and finally, by the preceding operator continuity, to (η, δ). We have proved

$$\mathcal{E}(F) = \mathcal{D}[f] = \tfrac{1}{4}\|L + L^*\|_{\mathcal{S}_2}^2 \geq 0. \tag{60}$$

Together with (50) this proves Theorem 1.2.

Proof of Theorem 1.1. The upper comparison $M_n \leq I_n(X_n)$ is Proposition 2.2. For the reverse asymptotic, a minimizer exists: in an orthonormal polynomial basis the objective is $\operatorname{tr}((E^*E)^{-1})$, which diverges when two nodes coalesce; compactness of the closed ordered node simplex then gives attainment. For large n the minimum is less than 2. The endpoint expansion in Lemma 3.1 makes a minimizing array contain both endpoints. Apply Theorem 1.2 to these arrays to get $\liminf_n n(M_n - I_n(X_n)) \geq 0$. This and (9) imply (2). □

Corollary 6.3. *If endpoint-containing arrays satisfy $n(I_n(Y_n)-I_n(X_n)) \to 0$, then their ordered angular displacement steps $n(\phi_i-\theta_i)$ tend to zero in every finite L^p on compact interior intervals.*

Proof. Every extracted limit has $\mathcal{D}[f] = 0$, so positivity of the kernel in (3) gives f/g constant almost everywhere. Outer normalization (27) fixes $F = (1-z^2)^{1/4}$ and therefore $\delta = 0$. Lemma 4.2 proves the assertion along every subsequence, which proves it for the full sequence. □

The supplementary directory `verification/` contains independent finite identity checks, the numerical reproduction programs and their outputs, and instructions for repeating them. In particular, they check the cardinal mass identity, the rank-one commutator, the exterior-resolvent factorization, (25), (35), and a global smooth-phase comparison. These calculations provide error-detection and reproducibility; none is a hypothesis or a substitute for an analytic

limiting argument in the proof. The conclusion concerns the first-order asymptotic and the interior rigidity above, and does not determine the exact minimizing arrays for finite n.

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